Grid Voltage Behaviour During a Single Phase-to-Earth Fault: Phasor Analysis in the RST Plane

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Abstract

The single phase-to-earth fault is the most frequent fault type in three-phase networks, accounting for roughly 70–80 % of all faults. Unlike phase-to-phase faults, its effect on the network voltages is dominated by the neutral earthing method: the same fault produces a barely detectable neutral displacement in an isolated-neutral network and a deep voltage dip in a solidly earthed one. This note derives the phase-to-earth voltages during a bolted and a resistive single phase-to-earth fault using symmetrical components, introduces the earth fault factor, and illustrates the results with phasor diagrams in the RST plane for the isolated, resonant and effectively earthed cases. The displacement locus of the neutral point as a function of the fault resistance is also given, together with the propagation of the fault through a Dy11 transformer and a short discussion of the consequences for protection and for grid-connected power converters.

1 Introduction

Consider a three-phase network with phases R, S, T and a fault between phase R and earth at a location F. The quantities of interest are the three phase-to-earth voltages $\overline{U}_{\mathrm{R}}$, $\overline{U}_{\mathrm{S}}$, $\overline{U}_{\mathrm{T}}$ at the fault location and, by extension, at the terminals of any equipment referenced to earth. Before the fault the network is balanced:

$$\overline{E}_{\mathrm{R}} = E, \qquad \overline{E}_{\mathrm{S}} = a^2 E, \qquad \overline{E}_{\mathrm{T}} = aE, \qquad a = e^{j2\pi/3}, \tag{1}$$

where E is the phase-to-earth voltage magnitude, $E = U_{\mathrm{n}}/\sqrt{3}$ with U_{n} the nominal line-to-line voltage.

The qualitative behaviour during the fault can be summarised as follows and is derived quantitatively in the rest of the note: the faulted phase voltage to earth collapses (fully for a bolted fault, partially for a resistive one); the healthy phase voltages to earth rise, by an amount that depends on the zero-sequence impedance of the network, up to the full line-to-line value $\sqrt{3}\,E$ in isolated and resonant-earthed networks; a zero-sequence (residual) voltage appears, which is the standard detection quantity for earth-fault protection; and the line-to-line voltage between the two healthy phases, $\overline{U}_{\mathrm{ST}}$, is unaffected in every earthing configuration, while $\overline{U}_{\mathrm{RS}}$ and $\overline{U}_{\mathrm{RT}}$ shrink only to the extent that fault current actually flows.

2 Symmetrical-Component Analysis

2.1 Sequence network connection

For a single phase-to-earth fault on phase R through a fault impedance Z_{F}, the boundary conditions at the fault location are

$$\overline{I}_{\mathrm{S}} = \overline{I}_{\mathrm{T}} = 0, \qquad \overline{U}_{\mathrm{R}} = Z_{\mathrm{F}}\,\overline{I}_{\mathrm{R}}. \tag{2}$$

Transforming to sequence quantities shows that the three sequence currents are equal and that the three sequence networks are connected in series with $3Z_{\mathrm{F}}$ (Fig. 1):

$$\overline{I}_1 = \overline{I}_2 = \overline{I}_0 = \frac{\overline{E}_{\mathrm{R}}}{Z_1 + Z_2 + Z_0 + 3Z_{\mathrm{F}}}, \qquad \overline{I}_{\mathrm{F}} = \overline{I}_{\mathrm{R}} = 3\overline{I}_0, \tag{3}$$

where Z_1, Z_2, Z_0 are the positive-, negative- and zero-sequence Thevenin impedances of the network seen from F. For networks without large rotating machines close to the fault, $Z_2 \simeq Z_1$ is an excellent approximation and is used throughout this note.

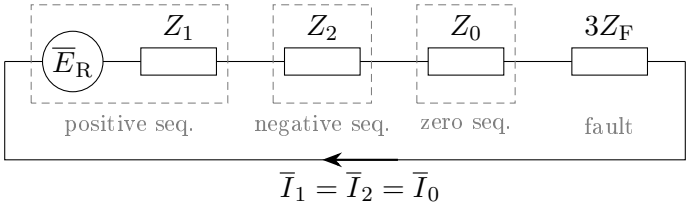


Figure 1: Series connection of the sequence networks for a single phase-to-earth fault on phase R through Z_{F}.

2.2 Phase-to-earth voltages at the fault

With $Z_2 = Z_1$ and a bolted fault ($Z_{\mathrm{F}} = 0$), the sequence voltages at the fault are $\overline{U}_1 = \overline{E}_{\mathrm{R}}(Z_0+Z_1)/(Z_0+2Z_1)$, $\overline{U}_2 = -\overline{E}_{\mathrm{R}}\, Z_1/(Z_0+2Z_1)$ and $\overline{U}_0 = -\overline{E}_{\mathrm{R}}\, Z_0/(Z_0+2Z_1)$. Transforming back to phase quantities gives the compact result

$$\boxed{\overline{U}_{\mathrm{R}} = 0, \qquad \overline{U}_{\mathrm{S}} = \left(a^2 - k\right)\overline{E}_{\mathrm{R}}, \qquad \overline{U}_{\mathrm{T}} = (a - k)\,\overline{E}_{\mathrm{R}}, \qquad k \triangleq \frac{Z_0 - Z_1}{Z_0 + 2Z_1}.} \tag{4}$$

The complex factor k contains the entire influence of the earthing method. Graphically, (4) states that the healthy-phase phasors keep their pre-fault orientation and are shifted rigidly by $-k\overline{E}_{\mathrm{R}}$, i.e. the effective earth reference point moves by $k\overline{E}_{\mathrm{R}}$ towards the faulted-phase vertex of the voltage triangle. The zero-sequence voltage, which is what residual-voltage relays measure, is

$$\overline{U}_0 = \tfrac{1}{3}\left(\overline{U}_{\mathrm{R}} + \overline{U}_{\mathrm{S}} + \overline{U}_{\mathrm{T}}\right) = -\frac{Z_0}{Z_0 + 2Z_1}\,\overline{E}_{\mathrm{R}}. \tag{5}$$

Two invariants are worth noting. First, $\overline{U}_{\mathrm{ST}} = \overline{U}_{\mathrm{S}} - \overline{U}_{\mathrm{T}} = (a^2 - a)\overline{E}_{\mathrm{R}}$ is identical to its pre-fault value for any k: the line-to-line voltage between the healthy phases never changes. Second, the line-to-line voltages involving the faulted phase, $\overline{U}_{\mathrm{RS}}$ and $\overline{U}_{\mathrm{RT}}$, have magnitude $|k - a^2|\,E$ and are reduced only when significant fault current flows ($k \to 0$); in an isolated network ($k \to 1$) the whole line-to-line triangle is preserved.

2.3 Earth fault factor

The earth fault factor k_{e} (IEC 60071-1) is the ratio between the highest healthy phase-to-earth voltage during the fault and the pre-fault phase-to-earth voltage at the same location. From (4), for real k,

$$k_{\mathrm{e}} = \left|a^2 - k\right| = \sqrt{k^2 + k + 1}. \tag{6}$$

For purely inductive impedances with $m = X_0/X_1$, $k = (m-1)/(m+2)$. Table 1 gives representative values. A system is called *effectively earthed* when $X_0/X_1 \le 3$ and $R_0/X_1 \le 1$, which bounds k_{e} to about 1.4; insulation, surge arresters and equipment withstand are coordinated on this basis.

Table 1: Earth fault factor versus zero-to-positive sequence reactance ratio (purely inductive network, bolted fault, $Z_2 = Z_1$).

X_0/X_1	k	k_e
1	0	1.00
2	0.25	1.15
3	0.40	1.25
5	0.57	1.38
10	0.75	1.52
$\to \infty$	$\to 1$	$\to \sqrt{3} \approx 1.73$

In an isolated-neutral network the zero-sequence impedance is capacitive, $Z_0 \simeq 1/(j\omega C_\mathrm{E})$ per phase with C_E the phase-to-earth capacitance, so that $k = (X_{C0} + X_1)/(X_{C0} - 2X_1)$ is slightly *greater* than one and the healthy phases can exceed $\sqrt{3}\,E$ by a few percent even in steady state. Larger overvoltages in isolated networks come from the transient phenomena discussed in Section 6.

3 Behaviour by Earthing Method

3.1 Isolated neutral

With the neutral isolated, the only zero-sequence path is the phase-to-earth capacitance and $|Z_0| \gg |Z_1|$, hence $k \simeq 1$. The picture is the classic one of Fig. 3: the line-to-line voltage triangle RST is unchanged, and the earth reference point moves from the centroid N to the faulted-phase vertex R. Consequently:

$$\overline{U}_\mathrm{R} = 0, \qquad |\overline{U}_\mathrm{S}| = |\overline{U}_\mathrm{T}| = \sqrt{3}\,E, \qquad \overline{U}_0 = -\overline{E}_\mathrm{R}, \tag{7}$$

with an angle of 60° between $\overline{U}_\mathrm{S}$ and $\overline{U}_\mathrm{T}$ instead of the pre-fault 120°. The residual voltage reaches its maximum possible value, $|3\overline{U}_0| = 3E$, which makes detection trivial, while the fault current is only the capacitive current of the two healthy phases,

$$|\overline{I}_\mathrm{F}| \simeq 3\,\omega C_\mathrm{E}\,E = \sqrt{3}\,\omega C_\mathrm{E}\,U_\mathrm{n}, \tag{8}$$

typically a few amperes to a few tens of amperes in a medium-voltage cable network. Loads and converters connected line-to-line are unaffected, and operation can continue during the fault; the price is that the entire network insulation must withstand line-to-line voltage to earth for the duration of the fault, and that a second earth fault on another phase (cross-country fault) short-circuits two phases through earth.

3.2 Resonant earthing (Petersen coil)

With an arc-suppression coil L_P between neutral and earth tuned to the network capacitance, $3\omega^2 L_\mathrm{P} C_\mathrm{E} \simeq 1$, the inductive neutral current cancels the capacitive fault current, leaving only a small resistive and harmonic residual. The steady-state voltage picture is the same as in the isolated case ($k \simeq 1$, healthy phases at $\sqrt{3}\,E$, full neutral displacement), but the near-zero current at the fault point allows arc self-extinction in air-insulated networks and permits sustained operation with the fault present. This is the standard practice in most European medium-voltage networks, including the Italian ones, which migrated from isolated to compensated neutral in the 2000s.

3.3 Effective and solid earthing

With the neutral solidly earthed or earthed through a low impedance, Z_0 is comparable to Z_1, k is small, and the fault becomes a true short circuit: the faulted phase voltage collapses over a wide area, the healthy phases rise moderately ($k_\mathrm{e} \le 1.4$ for effective earthing), and the fault current is of the same order as the three-phase short-circuit current. Fast selective tripping is mandatory. This is the usual arrangement in transmission networks and in low-voltage TN systems. From the point of view of connected equipment the event is a single-phase voltage dip with moderate swell on the healthy phases, rather than a neutral displacement.

4 Phasor Diagrams in the RST Plane

All diagrams are drawn in a common frame with the earth potential G at the origin, phase R pointing upwards, and voltages in per unit of E. Dashed grey phasors are the pre-fault quantities, thick black phasors the quantities during the fault.

Fig. 2 shows the balanced pre-fault condition: the star of $\overline{E}_\mathrm{R}$, $\overline{E}_\mathrm{S}$, $\overline{E}_\mathrm{T}$ with the neutral point N coincident with earth G, and the line-to-line triangle RST.

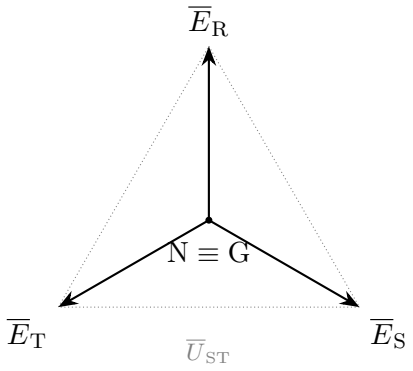


Figure 2: Balanced pre-fault condition: neutral point N at earth potential, symmetric star of phase-to-earth voltages, line-to-line triangle RST (dotted).

Fig. 3 shows the bolted fault on phase R in an isolated-neutral (or resonant-earthed) network, $k = 1$. The triangle RST is rigid and simply translates by $-\overline{E}_\mathrm{R}$: vertex R lands on the earth point G, the neutral N moves to $-\overline{E}_\mathrm{R}$, and the phase-to-earth voltages of the healthy phases become the line-to-line phasors $\overline{U}'_\mathrm{S} = \overline{U}_\mathrm{SR}$ and $\overline{U}'_\mathrm{T} = \overline{U}_\mathrm{TR}$, of magnitude $\sqrt{3}\,E$ and 60° apart.

Fig. 4 shows the same fault in an effectively earthed network with $X_0/X_1 = 3$, i.e. $k = 0.4$. The healthy-phase phasors shift by $-0.4\,\overline{E}_\mathrm{R}$, giving $k_\mathrm{e} = 1.25$. The triangle is no longer rigid: R collapses onto G, so $\overline{U}_\mathrm{RS}$ and $\overline{U}_\mathrm{RT}$ shrink from $\sqrt{3}\,E$ to $1.25\,E$, while the base $\overline{U}_\mathrm{ST}$ is unchanged. The zero-sequence voltage is $\overline{U}_0 = -0.6\,\overline{E}_\mathrm{R}$.

5 Fault Resistance and Neutral Displacement Locus

Real earth faults are rarely bolted: tree contact, surface arcing and cable sheath faults present resistances from a few ohms to tens of kΩ. In an isolated network, neglecting the series impedance Z_1 against the capacitive Z_0, the neutral displacement for a fault through R_F is

$$\overline{U}_\mathrm{N} = -\,\frac{\overline{E}_\mathrm{R}}{1 + j\,3\omega C_\mathrm{E} R_\mathrm{F}}. \tag{9}$$

As R_F sweeps from 0 to ∞, (9) is a bilinear map of a real parameter and $\overline{U}_\mathrm{N}$ travels along a semicircle of diameter E between $-\overline{E}_\mathrm{R}$ and the origin (Fig. 5). The practical consequence for

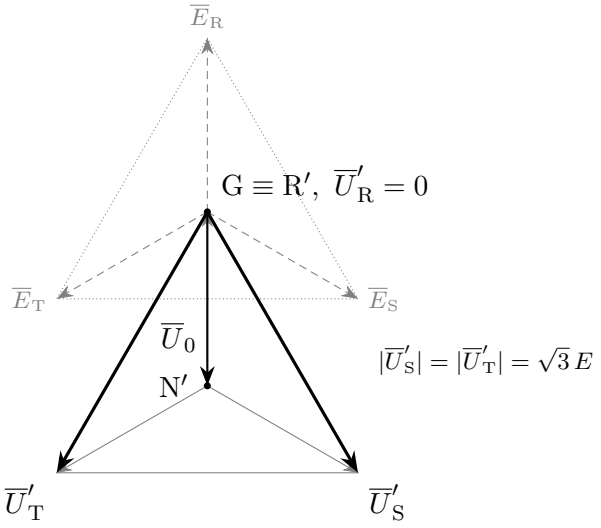


Figure 3: Bolted fault on phase R, isolated or resonant-earthed neutral ($k = 1$). The rigid voltage triangle translates so that vertex R coincides with earth: the neutral displaces fully ($\overline{U}_0 = -\overline{E}_\mathrm{R}$) and the healthy phases rise to line-to-line magnitude, 60° apart.

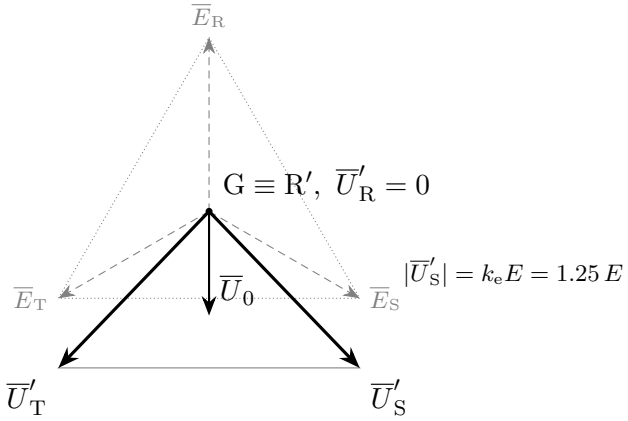


Figure 4: Bolted fault on phase R, effectively earthed network with $X_0/X_1 = 3$ ($k = 0.4$). The healthy phases rise only to $k_\mathrm{e} = 1.25$, the triangle distorts ($\overline{U}_\mathrm{RS}$, $\overline{U}_\mathrm{RT}$ reduced, $\overline{U}_\mathrm{ST}$ unchanged) and the neutral displacement is partial, $\overline{U}_0 = -0.6\,\overline{E}_\mathrm{R}$.

protection is that the residual voltage $|3\overline{U}_0|$ decreases monotonically with R_F: a relay set at, say, $0.3\,E$ residual (10 % of $3E$) detects faults only up to $R_\mathrm{F} \simeq \sqrt{10}/(\omega\,3C_\mathrm{E})$, and high-resistance faults require sensitive directional (wattmetric) methods based on $\overline{U}_0$ and $\overline{I}_0$ together.

6 Transient Phenomena

The phasor picture describes the steady state reached a few cycles after fault inception. The inception itself is accompanied by a high-frequency transient in which the healthy-phase capacitances recharge from phase to line-to-line voltage through the network series inductance, producing an oscillatory overshoot on the healthy phases well above $\sqrt{3}\,E$. In isolated networks, an unstable arc that extinguishes and restrikes near the current zeros (arcing ground fault) can pump this mechanism repeatedly and build phase-to-earth overvoltages historically quoted up to about 3 – 3.5 per unit; this, together with the cross-country fault risk of Section 3.1, is the main argument for resonant earthing or for adding a neutral resistor. In effectively earthed networks

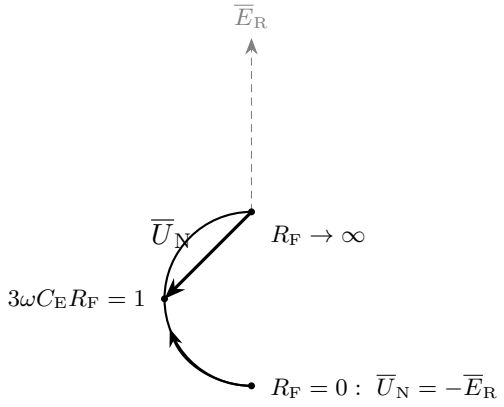


Figure 5: Locus of the neutral displacement $\overline{U}_\mathrm{N}$ in an isolated-neutral network as the fault resistance R_F varies from 0 (full displacement) to ∞ (no displacement). The arrow indicates increasing R_F.

the transient is a conventional short-circuit transient with DC offset in the fault current, and temporary overvoltages remain bounded by k_e.

7 Propagation Through a Dy11 Transformer

7.1 Sequence transformation rules

A transformer of vector group Dy11 (delta primary, wye secondary, clock number 11) transfers the sequence voltages between its terminals as, in per unit,

$$\overline{U}'_1 = e^{+j30^\circ}\,\overline{U}_1, \qquad \overline{U}'_2 = e^{-j30^\circ}\,\overline{U}_2, \qquad \overline{U}'_0 = 0 : \tag{10}$$

the positive sequence is rotated by $+30^\circ$ (secondary leads), the negative sequence by -30°, and the zero sequence is not transferred at all. The delta winding is excited only by the line-to-line voltages of the primary network, so the neutral displacement of Section 3 never reaches the secondary: the potential of the secondary star point is fixed by its own earthing, independently of the primary fault. Whether the secondary neutral is earthed (Dyn11) or not is irrelevant for a primary-side fault, since no secondary zero sequence is excited.

7.2 Secondary voltages for a primary phase-to-earth fault

Applying (10) to the sequence voltages of Section 2, with secondary phases labelled r, s, t, gives for the bolted fault on primary phase R (with $Z_2 = Z_1$):

$$\overline{U}'_\mathrm{s} = a^2 e^{+j30^\circ}\overline{U}_1 + a\,e^{-j30^\circ}\overline{U}_2 = -j\,\overline{E}_\mathrm{R}, \tag{11}$$

i.e. one secondary phase retains *exactly* its pre-fault value — in magnitude and in phase — for every earthing method and even for a resistive fault, because $a^2 e^{+j30^\circ} = -j$ and $a\,e^{-j30^\circ} = +j$ cancel the equal series drops of the positive and negative sequence. The other two phases dip symmetrically:

$$|\overline{U}'_\mathrm{r}| = |\overline{U}'_\mathrm{t}| = \frac{\sqrt{3 + (1+2k)^2}}{2\sqrt{3}}\,E, \qquad \Delta\varphi = \arctan\frac{\sqrt{3}}{1+2k} - 30^\circ, \tag{12}$$

with $\overline{U}'_\mathrm{r}$ advanced and $\overline{U}'_\mathrm{t}$ retarded by $\Delta\varphi$, so that the two phasors close towards each other. Numerically: for $k = 1$ (isolated or resonant primary neutral) the secondary is completely

unaffected, consistent with the rigid line-to-line triangle of Fig. 3; for $k = 0.4$ ($X_0/X_1 = 3$) the two phases drop to $0.72\,E$ with $\Delta\varphi \approx 14°$; for the deepest case $k = 0$ ($Z_0 = Z_1$) they drop to $E/\sqrt{3} \approx 0.58\,E$ with $\Delta\varphi = 30°$ (Fig. 6). In the Bollen classification the transformer converts the type B dip (single phase to earth, with zero sequence) into a type C dip (two phases reduced and closed towards each other, one phase intact). Two practical consequences follow: downstream of a Dy11 no phase-to-earth voltage ever drops below $1/\sqrt{3}$ of nominal for a single phase-to-earth fault, and one phase is always spared — which phase depends on the faulted primary phase and on the clock number. A fault impedance $Z_\mathrm{F} > 0$ only makes the dip shallower, while (11) continues to hold exactly.

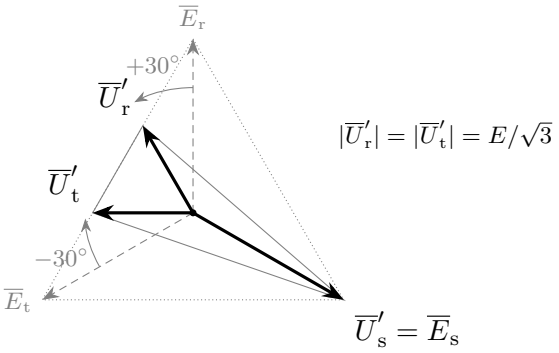


Figure 6: Secondary phase-to-earth voltages downstream of a Dy11 transformer for a bolted phase-to-earth fault on primary phase R, drawn for $k = 0$ ($Z_0 = Z_1$) in the secondary frame (the fixed 30° group displacement is omitted for clarity). Phase s is unaffected for every k; phases r and t drop to $E/\sqrt{3}$ and jump 30° towards each other (dip type C). For $k \to 1$ all three secondary voltages are unchanged.

7.3 Secondary-side earth fault seen from the primary

The blocking works in both directions. For an earth fault on the earthed-wye secondary side, the zero-sequence current circulates inside the delta winding and never appears in the primary lines: with sequence current $\overline{I}$ in the fault loop, the secondary line currents are $(3\overline{I},\, 0,\, 0)$ while the primary line currents are $(\sqrt{3}\overline{I},\, \sqrt{3}\overline{I},\, 0)$ in per unit — two primary lines carry equal current and the residual (zero-sequence) current on the primary side is identically zero. Seen from the primary network the fault therefore looks like a phase-to-phase event, and primary residual-current earth-fault protection is structurally blind to it: secondary earth faults must be cleared by secondary-side protection or by a relay in the transformer star-point connection.

8 Impact on Grid-Connected Converters

For grid-connected converters the relevant points are these. A three-wire converter without earthed midpoint provides no zero-sequence current path, so the fault affects it only through the line-to-line voltages and through the common-mode potential of the whole DC link, which follows the neutral displacement and stresses the phase-to-earth insulation, EMI filter Y capacitors and any earth-leakage monitoring. Synchronisation should be performed on the positive-sequence component (e.g. DDSRF- or DSOGI-based PLL): a per-phase or zero-sequence-sensitive PLL sees the full neutral displacement and can lose lock even though the line-to-line voltages are intact. In effectively earthed networks the fault appears as an asymmetric dip: the negative-sequence voltage drives double-frequency power and DC-link ripple, and ride-through with controlled positive- and negative-sequence current injection is required by most grid codes. Finally, converters with

an earthed DC midpoint or four-wire output do provide a zero-sequence path and must limit the resulting current actively during the neutral displacement.

9 Summary

The single phase-to-earth fault moves the network voltages in a way that is entirely governed by the ratio Z_0/Z_1, condensed in the factor $k = (Z_0 - Z_1)/(Z_0 + 2Z_1)$: the faulted phase-to-earth voltage collapses, the healthy ones become $(a^2 - k)\overline{E}_\mathrm{R}$ and $(a - k)\overline{E}_\mathrm{R}$, and a zero-sequence voltage $-\overline{E}_\mathrm{R} Z_0/(Z_0 + 2Z_1)$ appears. In isolated and resonant-earthed networks ($k \simeq 1$) the line-to-line triangle is rigid, the earth point moves to the faulted vertex and the healthy phases reach $\sqrt{3}\,E$; in effectively earthed networks (k small) the event is a single-phase short circuit with moderate rise of the healthy phases, $k_\mathrm{e} \leq 1.4$. The line-to-line voltage between the healthy phases is invariant in all cases, which is why line-to-line connected equipment, and converters synchronised on the positive sequence, ride through the disturbance naturally. Downstream of a Dy11 transformer the fault loses its zero sequence entirely: one secondary phase is exactly unaffected and the other two drop at most to $E/\sqrt{3}$ with $\pm 30°$ phase jumps towards each other (type C dip), while a secondary-side earth fault produces no residual current on the primary lines.

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